Assignment No	**12**
Name	**Satish Rathod**
Title	**Application of Linked Lists - Addition of Polynomial**
Objective	**Polynomial Creation**
Roll No	**MCA2649**

Source Code:-

```
//Name: Satish Rathod
//Roll No:MCA2649
//Module 4: Linked Lists
// Application of Linked Lists - Addition of Polynomial

import java.util.*;
//Node Template
class PNode
{
        int coeff;
        int exp;
        PNode next;

        PNode(int c,int e)
        {
```

```
            this.coeff = c;
            this.exp= e;
            this.next = null;
        }
}//end of pnode

//List Template
class Polynomial
{
        PNode head;

        //Insert a node in order of exponent
        void Insert(int coeff,int exp)
        {
            PNode t = new PNode(coeff,exp);

            //If the list is empty or exponent is greater than head
            if(head==null || exp>head.exp)
            {
                t.next = head;
                head = t;
                return;
            }

            //Traversal
```

```
        PNode temp = head;
        PNode prev = null;

        while(temp!=null && temp.exp>exp)
        {
                prev = temp;
                temp = temp.next;
        }
        prev.next = t;
        t.next = temp;


}//end of Insert

//Display the polynomial
void Display()
{
        if(head == null)
        {
                System.out.println("Empty Polynomial");
                return;
        }

        PNode temp = head;
        while(temp!=null)
```

```
            {
                System.out.print(temp.coeff +"x^"+temp.exp);
                if(temp.next!=null)
                {
                    System.out.print(" + ");
                }
                temp = temp.next;
            }
        }//end of display

}//end of Polynomial

class AddP
{
        public static void main(String args[])
        {
            Scanner sc = new Scanner(System.in);

            Polynomial p1 = new Polynomial();
            Polynomial p2 = new Polynomial();

            System.out.println("Enter the number of terms in the first
Polynomial. ");
            int n1 = sc.nextInt();
            System.out.println("Enter each terms as: coefficient exponent ");
```

```
            for(int i=0;i<n1;i++)
            {
                int c = sc.nextInt();
                int e = sc.nextInt();
                p1.Insert(c,e);
            }

            System.out.println("Enter the number of terms in the first
Polynomial. ");
            int n2 = sc.nextInt();
            System.out.println("Enter each terms as: coefficient exponent ");

            for(int i=0;i<n2;i++)
            {
                int c = sc.nextInt();
                int e = sc.nextInt();
                p2.Insert(c,e);
            }

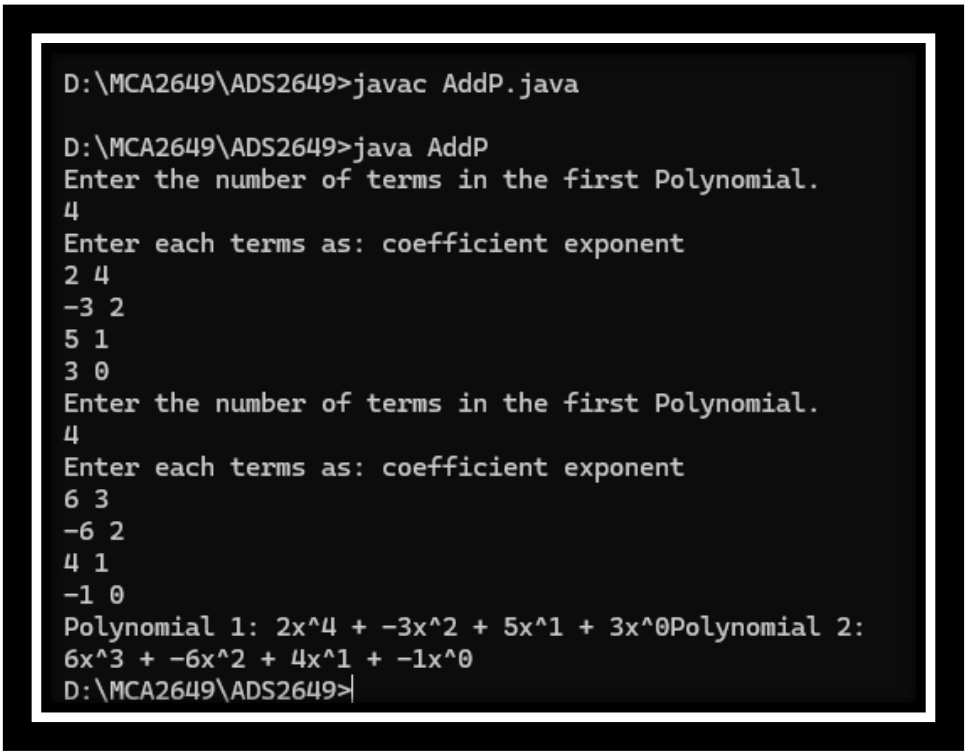
            System.out.print("Polynomial 1: ");
            p1.Display();

            System.out.println("Polynomial 2: ");
            p2.Display();

        }//end of the main
```

}

Output:-


D:\MCA2649\ADS2649>javac AddP.java
D:\MCA2649\ADS2649>java AddP
Enter the number of terms in the first Polynomial.
4
Enter each terms as: coefficient exponent
2 4
-3 2
5 1
3 0
Enter the number of terms in the first Polynomial.
4
Enter each terms as: coefficient exponent
6 3
-6 2
4 1
-1 0
Polynomial 1: 2x^4 + -3x^2 + 5x^1 + 3x^0Polynomial 2:
6x^3 + -6x^2 + 4x^1 + -1x^0
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